

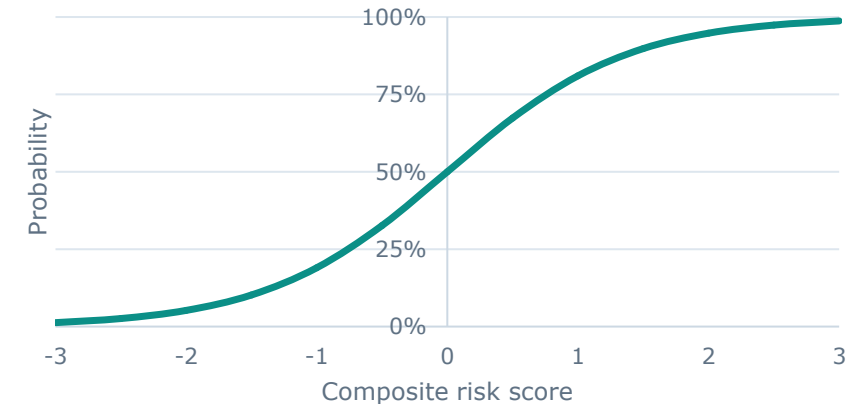
Logistic Regression- Predicting Loan Delinquencies

Business Case Study for hypothetical Bank



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Turn data into earlier, better decisions

Use historical patterns and current signals to estimate what is likely to happen next—then act before outcomes materialize.

01

SIGNALS

Combine behavior, profile and exposure data at a defined observation date.

02

PROBABILITY

Estimate an account-level likelihood for a specific future outcome.

03

ACTION

Prioritize review, monitoring or intervention while there is still time.

CASE STUDY APPLICATION

Predict 30-day delinquency risk for currently performing loan accounts.

Business problem

Identify currently performing accounts that may become delinquent within the next 30 days

THE DECISION QUESTION

Which current accounts deserve an earlier credit review?

Population

25,000 retail accounts with zero days past due on May 31, 2026

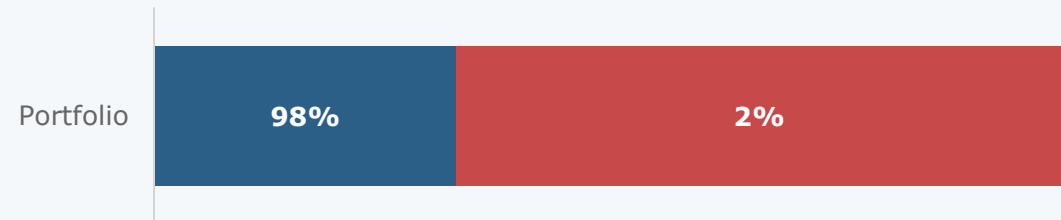
Observed outcome

500 accounts became delinquent by June 30, creating a 2% event rate

Analytical challenge

Class imbalance makes overall accuracy a weak measure of model quality

June outcome for accounts current in May



■ Remained current ■ Became delinquent

500

accounts became delinquent

2%

portfolio event rate

Solution approach

Separate model development from forward scoring to preserve a true future prediction



Timing logic

May characteristics explain June outcomes. After validation, June characteristics produce July probabilities. July actual performance later provides the out-of-time back-test.

Technology and data

A reproducible Python workflow converts monthly account snapshots into model-ready observations

DATA INPUTS

Observation snapshot

Account ID, days past due, FICO, exposure and internal credit grade

Performance snapshot

Next-month days past due joined by account ID to create the binary target

Model-ready dataset

One row per account with dated predictors and a future outcome

TECHNOLOGY STACK

Python

Model orchestration and scoring

pandas + NumPy

Joins, filters and feature preparation

scikit-learn

Scaling, logistic regression and validation

Matplotlib

Probability curves and evaluation visuals

VS Code

Development, versioning and repeatable execution

Leakage control

Every predictor must exist on the observation date. June DPD creates the June target and cannot appear among May predictors.

Model code and probability curve

Logistic regression converts combined credit characteristics into an account-level probability

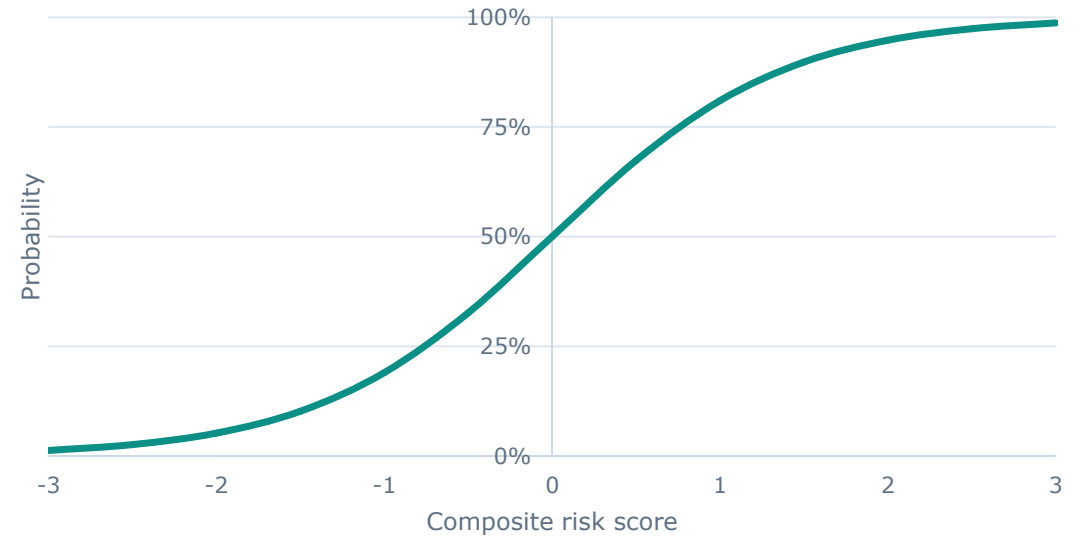
PYTHON MODEL CORE

```
# Build future outcome
df = may[may["DPD"] == 0].merge(
    june[["Account_ID", "DPD"]],
    on="Account_ID", how="left",
    suffixes=("_May", "_June"))
df["Y"] = (df["DPD_June"] > 0).astype(int)

# Fit and validate
features = ["FICO", "Exposure", "Rate"]
X_train, X_test, y_train, y_test = train_test_split(
    df[features], df["Y"], stratify=df["Y"], random_state=14)
scaler.fit(X_train)
model.fit(scaler.transform(X_train), y_train)

# Score current accounts
june_current["July_PD"] = model.predict_proba(
    scaler.transform(june_current[features]))[:, 1]
```

Illustrative logistic probability curve



A101

PD 42%

Actual outcome: delinquent

A105

PD 6%

Actual outcome: current

Findings and recommendations

The model narrows the review population and connects predictive risk with exposure-based prioritization

1,000

accounts flagged

At a 20% review threshold

60%

delinquencies captured

300 of 500 actual events

30%

precision

300 true positives among 1,000 alerts

EXPOSURE PRIORITIZATION

Account	Exposure	PD	Exposure × PD
B001	\$650K	46%	\$299K
B003	\$800K	27%	\$216K
B002	\$425K	34%	\$145K

Prioritization metric only. Expected credit loss also requires LGD and EAD aligned to the modeled event.

RECOMMENDED MANAGEMENT RESPONSE

Targeted review

Review the highest PD and exposure combinations first.

Credit action

Increase monitoring or adjust limits, terms and approvals when evidence supports action.

Model governance

Compare predicted and actual outcomes monthly, then recalibrate when performance shifts.